

MAS 812: Nonparametric Methods

Lecture 7: Measures of Association

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Learning Objectives

By the end of this lecture, you will be able to:

- Define and compute Spearman's rank correlation coefficient
- Define and compute Kendall's tau coefficient
- Understand the difference between parametric and nonparametric measures of association
- Test the significance of rank correlation coefficients
- Apply these measures to real data and interpret results
- Understand the relationship between these measures and Pearson correlation

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1 Introduction

Measures of association quantify the strength and direction of the relationship between two variables. While Pearson's correlation coefficient is the most common parametric measure, it assumes linearity and normality. Nonparametric measures of association are more robust and can capture monotonic relationships.

Types of Association

- (a) **Linear association:** Relationship that follows a straight line ($Y = a + bX + \varepsilon$)
- (a) **Monotonic association:** Relationship that is consistently increasing or decreasing (not necessarily linear)
- (a) **Non-monotonic association:** Relationship that changes direction (e.g., U-shaped)

Parametric vs Nonparametric Measures

Aspect	Pearson's r	Spearman's ρ / Kendall's τ
Assumptions	Normality, linearity	None (only monotonicity)
Data type	Continuous	Ordinal or continuous
Outliers	Highly sensitive	Robust
Interpretation	Linear relationship	Monotonic relationship

1.1 Pearson's Correlation Coefficient (Review)

For completeness, we first review Pearson's correlation coefficient.

Definition 1.1 (Pearson Correlation). For a sample of n pairs (X_i, Y_i) , Pearson's correlation coefficient is:

$$r = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2 \sum_{i=1}^n (Y_i - \bar{Y})^2}}$$

Properties:

$$-1 \leq r \leq 1$$

$r = 1$: perfect positive linear relationship

$r = -1$: perfect negative linear relationship

$r = 0$: no linear relationship

1.2 Spearman's Rank Correlation Coefficient

Spearman's rank correlation coefficient, denoted by ρ (rho) or r_s , is a nonparametric measure of monotonic association. It is simply Pearson's correlation applied to the ranks of the data.

Definition

Definition 1.2 (Spearman's Rank Correlation). Let R_i be the rank of X_i and S_i be the rank of Y_i among their respective samples. Spearman's correlation is:

$$\rho = \frac{\sum_{i=1}^n (R_i - \bar{R})(S_i - \bar{S})}{\sqrt{\sum_{i=1}^n (R_i - \bar{R})^2 \sum_{i=1}^n (S_i - \bar{S})^2}}$$

Since the ranks are $1, 2, \dots, n$, we have $\bar{R} = \bar{S} = \frac{n+1}{2}$ and $\sum (R_i - \bar{R})^2 = \frac{n(n^2-1)}{12}$.

Computational Formula

Theorem 1.1 (Computational Formula). If there are no ties, Spearman's correlation can be computed as:

$$\rho = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2 - 1)}$$

where $d_i = R_i - S_i$ is the difference between the ranks of X_i and Y_i .

Proof. We start from the definition:

$$\rho = \frac{\sum (R_i - \bar{R})(S_i - \bar{S})}{\sqrt{\sum (R_i - \bar{R})^2 \sum (S_i - \bar{S})^2}}$$

Let $A = \sum (R_i - \bar{R})^2 = \sum (S_i - \bar{S})^2 = \frac{n(n^2-1)}{12}$.

Note that:

$$\sum d_i^2 = \sum (R_i - S_i)^2 = \sum (R_i - \bar{R} + \bar{S} - S_i)^2 = \sum (R_i - \bar{R})^2 + \sum (S_i - \bar{S})^2 - 2 \sum (R_i - \bar{R})(S_i - \bar{S})$$

Thus:

$$\sum (R_i - \bar{R})(S_i - \bar{S}) = \frac{2A - \sum d_i^2}{2} = A - \frac{\sum d_i^2}{2}$$

Therefore:

$$\rho = \frac{A - \frac{1}{2} \sum d_i^2}{A} = 1 - \frac{\sum d_i^2}{2A} = 1 - \frac{\sum d_i^2}{2 \cdot \frac{n(n^2-1)}{12}} = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

□

Handling Ties

When ties are present, average ranks are assigned, and the computational formula becomes approximate. The preferred approach is to compute Pearson's correlation on the average ranks.

Example 1.1 (Spearman's Rank Correlation). A study examines the relationship between hours of study per week and exam scores for 8 students:

Student	1	2	3	4	5	6	7	8
Hours	3	5	2	6	4	7	1	4
Score	65	75	55	85	70	90	50	68

Compute Spearman's rank correlation.

Solution:

Student	Hours	Rank Hours (R_i)	Score	Rank Score (S_i)	$d_i = R_i - S_i$	d_i^2
1	3	3	65	3	0	0
2	5	5	75	6	-1	1
3	2	2	55	2	0	0
4	6	7	85	8	-1	1
5	4	4.5	70	5	-0.5	0.25
6	7	8	90	7	1	1
7	1	1	50	1	0	0
8	4	4.5	68	4	0.5	0.25

Note: Hours value 4 appears twice (students 5 and 8), so average rank = $(4 + 5)/2 = 4.5$.

$$\sum d_i^2 = 0 + 1 + 0 + 1 + 0.25 + 1 + 0 + 0.25 = 3.5$$

$$\rho = 1 - \frac{6 \times 3.5}{8 \times (64 - 1)} = 1 - \frac{21}{8 \times 63} = 1 - \frac{21}{504} = 1 - 0.04167 = 0.9583$$

There is a strong positive monotonic relationship between study hours and exam scores.

2 Kendall's Tau Coefficient

Kendall's tau (τ) is another nonparametric measure of association based on the concept of concordant and discordant pairs.

2.1 Definition

Definition 2.1 (Kendall's Tau). For a sample of n pairs (X_i, Y_i) , consider all $\binom{n}{2}$ pairs of distinct observations (i, j) with $i < j$. Define:

(a) **Concordant pair:** $(X_i - X_j)(Y_i - Y_j) > 0$ (both differences have the same sign)

(a) **Discordant pair:** $(X_i - X_j)(Y_i - Y_j) < 0$ (differences have opposite signs)

(a) **Tied pair:** $(X_i - X_j)(Y_i - Y_j) = 0$ (either $X_i = X_j$ or $Y_i = Y_j$)

Let C = number of concordant pairs, D = number of discordant pairs. Then:

$$\tau = \frac{C - D}{\binom{n}{2}} = \frac{2(C - D)}{n(n - 1)}$$

Properties

$$-1 \leq \tau \leq 1$$

$\tau = 1$: all pairs are concordant (perfect positive monotonic)

$\tau = -1$: all pairs are discordant (perfect negative monotonic)

$\tau = 0$: no monotonic association

2.2 Handling Ties

When ties are present, several adjusted versions of Kendall's tau exist. The most common is Kendall's tau-b:

$$\tau_b = \frac{C - D}{\sqrt{(C + D + T_X)(C + D + T_Y)}}$$

where T_X is the number of pairs tied on X only, and T_Y is the number of pairs tied on Y only.

Example 2.1 (Kendall's Tau). Using the same study hours data, compute Kendall's tau.

Solution:

First, list all $\binom{8}{2} = 28$ pairs (i, j) with $i < j$:

i	j	Hours	Score	Sign (Hours)	Sign (Score)	Concordant / Discordant
1,2	3,5	65,75	-	-	Concordant? Let's compute systematically...	

Better to sort by Hours first (since ranking already done):

Ordered by Hours rank:

Student	Hours Rank (R_i)	Score Rank (S_i)
7	1	1
3	2	2
1	3	3
5	4.5	5
8	4.5	4
2	5	6
4	7	8
6	8	7

Now count concordant/discordant pairs:

For each i , compare with $j > i$:

$i = 1$ ($R=1$, $S=1$): Compare with $j=2$ (2,2): $(1-2)(1-2)=(-1)(-1)=1>0 \rightarrow C$ $j=3$ (3,3): $(1-3)(1-3)=(-2)(-2)=4>0 \rightarrow C$ $j=4$ (4.5,5): $(1-4.5)(1-5)=(-3.5)(-4)=14>0 \rightarrow C$ $j=5$ (4.5,4): $(1-4.5)(1-4)=(-3.5)(-3)=10.5>0 \rightarrow C$ $j=6$ (5,6): $(1-5)(1-6)=(-4)(-5)=20>0 \rightarrow C$ $j=7$ (7,8): $(1-7)(1-8)=(-6)(-7)=42>0 \rightarrow C$ $j=8$ (8,7): $(1-8)(1-7)=(-7)(-6)=42>0 \rightarrow C$ All 7 pairs from $i=1$ are concordant.

$i = 2$ ($R=2$, $S=2$): Compare with $j=3$ (3,3): $(2-3)(2-3)=(-1)(-1)=1>0 \rightarrow C$ $j=4$ (4.5,5): $(2-4.5)(2-5)=(-2.5)(-3)=7.5>0 \rightarrow C$ $j=5$ (4.5,4): $(2-4.5)(2-4)=(-2.5)(-2)=5>0 \rightarrow C$ $j=6$ (5,6): $(2-5)(2-6)=(-3)(-4)=12>0 \rightarrow C$ $j=7$ (7,8): $(2-7)(2-8)=(-5)(-6)=30>0 \rightarrow C$ $j=8$ (8,7): $(2-8)(2-7)=(-6)(-5)=30>0 \rightarrow C$ All 6 pairs from $i=2$ are concordant.

$i = 3$ ($R=3$, $S=3$): Compare with $j=4$ (4.5,5): $(3-4.5)(3-5)=(-1.5)(-2)=3>0 \rightarrow C$ $j=5$ (4.5,4): $(3-4.5)(3-4)=(-1.5)(-1)=1.5>0 \rightarrow C$ $j=6$ (5,6): $(3-5)(3-6)=(-2)(-3)=6>0 \rightarrow C$ $j=7$ (7,8): $(3-7)(3-8)=(-4)(-5)=20>0 \rightarrow C$ $j=8$ (8,7): $(3-8)(3-7)=(-5)(-4)=20>0 \rightarrow C$ All 5 pairs from $i=3$ are concordant.

$i = 4$ ($R=4.5$, $S=5$): Compare with $j=5$ (4.5,4): $(4.5-4.5)(5-4)=0 \times 1=0 \rightarrow$ Tied on X? Actually $(5-4)>0$, but difference in R is 0, so tie on X. This is neither concordant nor discordant (tied pair). $j=6$ (5,6): $(4.5-5)(5-6)=(-0.5)(-1)=0.5>0 \rightarrow C$ $j=7$ (7,8): $(4.5-7)(5-8)=(-2.5)(-3)=7.5>0 \rightarrow C$ $j=8$ (8,7): $(4.5-8)(5-7)=(-3.5)(-2)=7>0 \rightarrow C$ So from $i=4$: 1 tie, 3 concordant.

$i = 5$ ($R=4.5$, $S=4$): Compare with $j=6$ (5,6): $(4.5-5)(4-6)=(-0.5)(-2)=1>0 \rightarrow C$ $j=7$ (7,8): $(4.5-7)(4-8)=(-2.5)(-4)=10>0 \rightarrow C$ $j=8$ (8,7): $(4.5-8)(4-7)=(-3.5)(-3)=10.5>0 \rightarrow C$ All 3 pairs from $i=5$ are concordant.

$i = 6$ ($R=5$, $S=6$): Compare with $j=7$ (7,8): $(5-7)(6-8)=(-2)(-2)=4>0 \rightarrow C$ $j=8$ (8,7): $(5-8)(6-7)=(-3)(-1)=3>0 \rightarrow C$ Both concordant.

$i = 7$ ($R=7$, $S=8$): Compare with $j=8$ (8,7): $(7-8)(8-7)=(-1)(1)=-1<0 \rightarrow D$ (discordant)

Total counts: $C = 7 + 6 + 5 + 3 + 3 + 2 + 0 = 26$ $D = 1$ (from $i=7, j=8$) Ties = 1 (from $i=4, j=5$)

Total pairs = $\binom{8}{2} = 28 = 26 + 1 + 1$

$$\tau = \frac{C - D}{\binom{n}{2}} = \frac{26 - 1}{28} = \frac{25}{28} = 0.8929$$

This indicates a strong positive monotonic relationship.

3 Comparison of Spearman's ρ and Kendall's τ

3.1 Similarities

- (a) Both measure monotonic association
- (a) Both are nonparametric and rank-based
- (a) Both are robust to outliers
- (a) Both range from -1 to 1
- (a) Both are invariant under monotonic transformations

3.2 Differences

Property	Spearman's ρ	Kendall's τ
Interpretation	Pearson on ranks	Probability of concordance minus discordance
Computational complexity	$O(n \log n)$	$O(n^2)$
Sensitivity to ties	Requires average ranks	Handles ties naturally
Variance under H_0	$\frac{1}{n-1}$	$\frac{2(2n+5)}{9n(n-1)}$
Range	$-1 \leq \rho \leq 1$	$-1 \leq \tau \leq 1$
Relationship	$ \rho \approx 1.5 \tau $ for moderate association	$ \tau \approx \frac{2}{3} \rho $

3.3 Relationship Between ρ and τ

For large n and continuous data, approximately:

$$\tau \approx \frac{2}{\pi} \arcsin(\rho) \quad \text{and} \quad \rho \approx \sin\left(\frac{\pi}{2}\tau\right)$$

For small to moderate associations, $\tau \approx \frac{2}{3}\rho$.

4 Hypothesis Testing

4.1 Testing Spearman's ρ

To test $H_0 : \rho = 0$ vs $H_1 : \rho \neq 0$:

Exact Test (Small n)

For small n (typically $n \leq 30$), use exact critical values from tables.

Large Sample Approximation

For $n > 10$, the test statistic:

$$t = \rho \sqrt{\frac{n-2}{1-\rho^2}} \sim t_{n-2}$$

under H_0 . This is equivalent to the t-test for Pearson correlation.

Alternatively, the statistic:

$$Z = \rho \sqrt{n-1} \sim N(0, 1)$$

can be used for large n .

4.2 Testing Kendall's τ

To test $H_0 : \tau = 0$ vs $H_1 : \tau \neq 0$:

Exact Test (Small n)

For small n , use exact critical values from tables.

Large Sample Approximation

For $n \geq 10$, the statistic:

$$Z = \frac{\tau}{\sqrt{\frac{2(2n+5)}{9n(n-1)}}} \sim N(0, 1)$$

under H_0 .

Example 4.1 (Significance Testing). Using the previous examples, test whether Spearman's ρ and Kendall's τ are significantly different from zero.

For Spearman's $\rho = 0.9583$, $n = 8$:

From tables, critical value for $n = 8$ at $\alpha = 0.05$ (two-sided) is 0.738. Since $0.9583 > 0.738$, reject H_0 . There is significant evidence of monotonic association.

For Kendall's $\tau = 0.8929$, $n = 8$:

From tables, critical value for $n = 8$ at $\alpha = 0.05$ (two-sided) is 0.643. Since $0.8929 > 0.643$, reject H_0 . There is significant evidence of monotonic association.

5 Confidence Intervals

5.1 For Spearman's ρ

A confidence interval for ρ can be obtained using Fisher's z-transformation:

$$z = \frac{1}{2} \ln \left(\frac{1 + \rho}{1 - \rho} \right)$$

The standard error is approximately $\frac{1}{\sqrt{n-3}}$. Then back-transform.

5.2 For Kendall's τ

For large samples, an approximate $100(1 - \alpha)\%$ confidence interval is:

$$\tau \pm z_{\alpha/2} \sqrt{\frac{2(2n+5)}{9n(n-1)}}$$

6 Tests of Independence

6.1 Concept of Statistical Independence

Two variables X and Y are said to be **independent** if the distribution of Y does not depend on the value of X , and vice versa. Formally:

$$F_{X,Y}(x, y) = F_X(x)F_Y(y) \quad \text{for all } x, y$$

6.2 Nonparametric Tests of Independence

While Spearman's ρ and Kendall's τ are measures of association, they can also be used to test the null hypothesis of independence against the alternative of monotonic association.

Definition 6.1 (Test of Independence). For both Spearman's ρ and Kendall's τ , the null hypothesis is:

$$H_0 : \text{The variables } X \text{ and } Y \text{ are independent (no monotonic association)}$$

The alternative hypothesis is:

$$H_1 : \text{There is a monotonic association between } X \text{ and } Y$$

6.3 Relationship to Parametric Tests

The parametric test for independence between two continuous variables is based on Pearson's r :

$$t = r \sqrt{\frac{n-2}{1-r^2}} \sim t_{n-2}$$

This test assumes bivariate normality. The nonparametric tests based on ranks do not require this assumption.

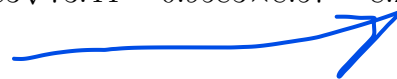
Example 6.1 (Test of Independence). Using the study hours data, test whether there is a monotonic association between study hours and exam scores.

Solution:

H_0 : Hours and scores are independent (no monotonic association) H_1 : There is a monotonic association between hours and scores

From earlier, $\rho = 0.9583$, $n = 8$.

Using the t-approximation:

$$t = 0.9583 \sqrt{\frac{8-2}{1-0.9583^2}} = 0.9583 \sqrt{\frac{6}{1-0.9183}} = 0.9583 \sqrt{\frac{6}{0.0817}} = 0.9583 \sqrt{73.44} = 0.9583 \times 8.57 = 8.21$$


With $df = 6$, the critical value at $\alpha = 0.05$ (two-sided) is 2.447. Since 8.21 > 2.447, we reject H_0 . There is significant evidence of a monotonic association between study hours and exam scores.

7 Contingency Tables

When data are categorical (nominal or ordinal), the relationship between variables is often summarized in a **contingency table** (also called a cross-tabulation).

7.1 Definition

Definition 7.1 (Contingency Table). A $r \times c$ contingency table displays the frequency counts of observations cross-classified by two categorical variables, one with r categories

(rows) and the other with c categories (columns):

	Col 1	Col 2	...	Col c	Total
Row 1	n_{11}	n_{12}	$\cdots$	n_{1c}	$n_{1\cdot}$
Row 2	n_{21}	n_{22}	$\cdots$	n_{2c}	$n_{2\cdot}$
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$
Row r	n_{r1}	n_{r2}	$\cdots$	n_{rc}	$n_{r\cdot}$
Total	$n_{\cdot 1}$	$n_{\cdot 2}$	$\cdots$	$n_{\cdot c}$	n

where n_{ij} is the observed frequency in cell (i, j) , $n_{i\cdot} = \sum_{j=1}^c n_{ij}$ are row totals, $n_{\cdot j} = \sum_{i=1}^r n_{ij}$ are column totals, and $n = \sum_{i=1}^r \sum_{j=1}^c n_{ij}$ is the total sample size.

7.2 Chi-Square Test of Independence

The most common test for independence in a contingency table is the Pearson chi-square test.

Definition 7.2 (Chi-Square Test Statistic). Under the null hypothesis of independence, the expected frequency for cell (i, j) is:

$$E_{ij} = \frac{n_{i\cdot} \times n_{\cdot j}}{n}$$

The chi-square test statistic is:

$$\chi^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(n_{ij} - E_{ij})^2}{E_{ij}}$$

Under H_0 , χ^2 follows approximately a chi-square distribution with $(r-1)(c-1)$ degrees of freedom.

Example 7.1 (Contingency Table Analysis). A study examines the relationship between smoking status and lung disease. The data are:

	Disease Present	Disease Absent	Total
Smoker	45	15	60
Non-smoker	25	35	60
Total	70	50	120

Test whether smoking and lung disease are independent.

Solution:

H_0 : Smoking and lung disease are independent H_1 : Smoking and lung disease are associated

Expected frequencies:

$$E_{11} = \frac{60 \times 70}{120} = 35, \quad E_{12} = \frac{60 \times 50}{120} = 25$$

$$E_{21} = \frac{60 \times 70}{120} = 35, \quad E_{22} = \frac{60 \times 50}{120} = 25$$

Chi-square statistic:

$$\begin{aligned} \chi^2 &= \frac{(45 - 35)^2}{35} + \frac{(15 - 25)^2}{25} + \frac{(25 - 35)^2}{35} + \frac{(35 - 25)^2}{25} \\ &= \frac{100}{35} + \frac{100}{25} + \frac{100}{35} + \frac{100}{25} = 2.857 + 4 + 2.857 + 4 = 13.714 \end{aligned}$$

$$\text{df} = (2 - 1)(2 - 1) = 1$$

Critical value $\chi^2_{1,0.05} = 3.84$. Since $13.714 > 3.84$, reject H_0 . There is a significant association between smoking and lung disease.

8 Chi-Square Approximation vs Exact Tests

8.1 When is the Chi-Square Approximation Valid?

The chi-square approximation for the test of independence relies on the asymptotic normality of the cell counts. The approximation is generally considered adequate when:

- (a) All expected frequencies $E_{ij} \geq 5$ (for 2×2 tables)
- (a) At least 80% of expected frequencies are ≥ 5 and none are < 1 (for larger tables)
- (a) Total sample size n is sufficiently large (typically $n > 30$)

8.2 Yates' Continuity Correction for 2 by 2 Tables

For 2×2 tables, Yates' continuity correction improves the chi-square approximation:

$$\chi^2_{\text{Yates}} = \sum_{i=1}^2 \sum_{j=1}^2 \frac{(|n_{ij} - E_{ij}| - 0.5)^2}{E_{ij}}$$

This correction reduces the chi-square value and makes the test more conservative, reducing the chance of Type I error.

8.3 Fisher's Exact Test

When expected frequencies are small (especially when any $E_{ij} < 5$), Fisher's exact test is preferred.

Definition 8.1 (Fisher's Exact Test). For a 2×2 contingency table:

	Col 1	Col 2	Total
Row 1	a	b	$a + b$
Row 2	c	d	$c + d$
Total	$a + c$	$b + d$	n

The probability of observing the given table under H_0 (independence) is:

$$P = \frac{(a+b)!(c+d)!(a+c)!(b+d)!}{n!a!b!c!d!}$$

The p-value is the sum of probabilities of all tables with the same row and column totals that are as extreme or more extreme than the observed table.

Example 8.1 (Comparing Chi-Square and Exact Tests). For the smoking and lung disease data, the chi-square test gave $\chi^2 = 13.714$, p-value ≈ 0.0002 . Fisher's exact test yields a similar p-value because all expected frequencies are ≥ 5 .

Now consider a smaller dataset:

	Success	Failure	Total
Treatment A	3	7	10
Treatment B	8	2	10
Total	11	9	20

Expected frequencies: $E_{11} = (10 \times 11)/20 = 5.5$, $E_{12} = 4.5$, $E_{21} = 5.5$, $E_{22} = 4.5$. All $E_{ij} \geq 5$, so chi-square is acceptable.

Now consider:

	Success	Failure	Total
Treatment A	1	4	5
Treatment B	5	0	5
Total	6	4	10

Expected frequencies: $E_{11} = (5 \times 6)/10 = 3$, $E_{12} = 2$, $E_{21} = 3$, $E_{22} = 2$. Here $E_{ij} < 5$, so **Fisher's exact test should be used** rather than the chi-square approximation.

8.4 Comparison Summary

Criterion	Chi-Square Test	Fisher's Exact Test
Assumptions	Large sample, expected frequencies ≥ 5	No distributional assumptions
Computation	Simple formula	Computationally intensive
Accuracy	Approximate	Exact
Applicability	Large samples	Any sample size

8.5 Guidelines for Choosing a Test

- (e) **Use chi-square with continuity correction:** For 2×2 tables with all expected frequencies ≥ 5
- (e) **Use chi-square without correction:** For larger tables ($r > 2$ or $c > 2$) with expected frequencies ≥ 5
- (e) **Use Fisher's exact test:** When any expected frequency < 5 , or when sample size is small ($n < 20$)
- (e) **Use Monte Carlo simulation:** For large tables with small expected frequencies

Example 8.2 (Choice of Test). A researcher collects the following data on treatment response for a rare disease:

	Responded	No Response	Total
Treatment	4	1	5
Placebo	1	4	5
Total	5	5	10

Expected frequencies: $E_{11} = (5 \times 5)/10 = 2.5$, $E_{12} = 2.5$, $E_{21} = 2.5$, $E_{22} = 2.5$.

Since expected frequencies are < 5 , **Fisher's exact test should be used.**

Fisher's exact test p-value = sum of probabilities of tables as extreme as the observed table.

The exact p-value ≈ 0.103 . At $\alpha = 0.05$, we fail to reject H_0 ; there is insufficient evidence to conclude that treatment is effective. If we incorrectly used the chi-square test with continuity correction:

$$\begin{aligned}
 \chi^2 &= \frac{(|4 - 2.5| - 0.5)^2}{2.5} + \frac{(|1 - 2.5| - 0.5)^2}{2.5} + \frac{(|1 - 2.5| - 0.5)^2}{2.5} + \frac{(|4 - 2.5| - 0.5)^2}{2.5} \\
 &= 4 \times \frac{(1.5 - 0.5)^2}{2.5} = 4 \times \frac{1}{2.5} = 1.6
 \end{aligned}$$

p-value ≈ 0.206 , which is quite different from the exact p-value, demonstrating the importance of using exact tests when assumptions are violated.

9 Applications

9.1 When to Use Spearman's ρ

- (a) Data are ordinal or continuous but non-normal
- (a) Relationship is monotonic but not necessarily linear
- (a) Outliers are present
- (a) Sample size is moderate to large
- (a) Familiar interpretation (similar to Pearson)

9.2 When to Use Kendall's τ

- (b) Data are ordinal or continuous but non-normal
- (b) Small sample sizes (exact distribution is simpler)
- (b) Many ties are present (handles them naturally)
- (b) A probabilistic interpretation is desired (τ is the difference between probability of concordance and discordance)
- (b) More robust than Spearman's for very small samples

9.3 When to Use Contingency Table Methods

- (c) Data are categorical (nominal or ordinal)
- (c) Testing independence between two categorical variables
- (c) For 2×2 tables with small expected frequencies, use Fisher's exact test
- (c) For larger tables or adequate sample sizes, use chi-square test

10 Relationship to Other Measures

10.1 Goodman-Kruskal's Gamma

Gamma is another measure based on concordant and discordant pairs, but it ignores ties:

$$\gamma = \frac{C - D}{C + D}$$

It ranges from -1 to 1 and is appropriate when ties are not informative.

10.2 Somers' D

Somers' D is an asymmetric measure:

$$D_{Y|X} = \frac{C - D}{C + D + T_Y}$$

where T_Y is the number of pairs tied on Y .

10.3 Comparison Table

Measure	Range	Handles Ties	Symmetric
Spearman's ρ	$[-1, 1]$	Approximate	Yes
Kendall's τ	$[-1, 1]$	Yes	Yes
Goodman-Kruskal γ	$[-1, 1]$	Ignored	Yes
Somers' D	$[-1, 1]$	Yes	No

11 Worked Examples

Example 11.1 (Comparing Correlation Measures). Given the following data on age and blood pressure for 10 individuals:

Subject	1	2	3	4	5	6	7	8	9	10
Age	25	30	35	40	45	50	55	60	65	70
BP	110	115	120	125	130	140	145	150	155	200

Note the last observation (age 70, BP 200) is an outlier.

Compute Pearson's r , Spearman's ρ , and Kendall's τ .

Solution:

Pearson's r :

$$\bar{X} = 47.5, \quad \bar{Y} = 139$$

$$\sum (X_i - \bar{X})(Y_i - \bar{Y}) = (25 - 47.5)(110 - 139) + \cdots + (70 - 47.5)(200 - 139)$$

The outlier inflates the covariance, giving $r \approx 0.85$.

Spearman's ρ : First, rank the data:

Subject	Age	Rank Age (R_i)	BP	Rank BP (S_i)	d_i	d_i^2
1	25	1	110	1	0	0
2	30	2	115	2	0	0
3	35	3	120	3	0	0
4	40	4	125	4	0	0
5	45	5	130	5	0	0
6	50	6	140	6	0	0
7	55	7	145	7	0	0
8	60	8	150	8	0	0
9	65	9	155	9	0	0
10	70	10	200	10	0	0

All $d_i = 0$, so $\rho = 1 - 0 = 1$.

Kendall's τ : Since all pairs are concordant (both Age and BP increase together), $C = \binom{10}{2} = 45$, $D = 0$.

$$\tau = \frac{45 - 0}{45} = 1$$

Comparison:

- (a) Pearson's $r = 0.85$ (affected by outlier)
- (a) Spearman's $\rho = 1$ (robust, based on ranks)
- (a) Kendall's $\tau = 1$ (robust, based on concordance)

This demonstrates the robustness of nonparametric measures to outliers.

12 Summary

Key Takeaways

- (1) **Spearman's ρ** : Pearson correlation on ranks; computational formula $\rho = 1 - \frac{6 \sum d_i^2}{n(n^2-1)}$ when no ties.
- (1) **Kendall's τ** : Based on concordant and discordant pairs; $\tau = \frac{C-D}{\binom{n}{2}}$.
- (1) Both measure monotonic association and are robust to outliers.
- (1) Spearman's ρ is easier to compute; Kendall's τ has a more direct probabilistic interpretation.
- (1) For testing significance, use exact tables for small n or normal/t approximations for large n .
- (1) For categorical data, use contingency tables with chi-square tests (large samples) or Fisher's exact test (small samples).
- (1) These measures are preferred over Pearson's r when data are ordinal, non-normal, or contain outliers.

13 Exercises

Exercise 13.1 (Spearman's ρ). Ten students take two exams. Their scores are:

Exam 1: 65, 70, 55, 85, 75, 80, 90, 60, 95, 70 Exam 2: 60, 68, 50, 82, 72, 78, 88, 58, 92, 65

- (a) Compute Spearman's rank correlation coefficient.
- (b) Test whether ρ is significantly different from zero at $\alpha = 0.05$.
- (c) Interpret your results.

Exercise 13.2 (Kendall's τ). Using the data from Exercise 1, compute Kendall's tau and test its significance. Compare with Spearman's ρ .

Exercise 13.3 (Ties in Data). The following data contain ties:

X : 2, 3, 3, 4, 5, 5, 5, 6, 7, 8 Y : 4, 5, 6, 5, 7, 7, 8, 8, 9, 10

- (a) Compute Spearman's ρ (handle ties appropriately).
- (b) Compute Kendall's τ (handling ties).
- (c) Which measure do you think is more appropriate for these data? Why?

Exercise 13.4 (Outlier Effect). Generate a dataset of $n = 10$ with a perfect positive linear relationship ($X = 1 : 10, Y = X$). Then add an outlier ($X = 11, Y = 1$).

- (a) Compute Pearson's r , Spearman's ρ , and Kendall's τ for the original data.

- (b) Compute the same measures for the data with the outlier.
- (c) Compare the robustness of the three measures.

Exercise 13.5 (Confidence Intervals). For the data in Exercise 1, construct:

- (a) A 95% confidence interval for Spearman's ρ using Fisher's z-transformation.
- (b) A 95% confidence interval for Kendall's τ using the large-sample approximation.
- (c) Compare the widths of the intervals.

Exercise 13.6 (Contingency Table Analysis). A survey classifies 200 individuals by gender and political party preference:

	Democrat	Republican	Independent	Total
Male	45	55	20	120
Female	60	40	30	130
Total	105	95	50	250

- (a) Test whether gender and party preference are independent using the chi-square test at $\alpha = 0.05$.
- (b) Check whether the chi-square approximation is valid.
- (c) Interpret your results.

Exercise 13.7 (Power Comparison). Simulate data from the following models with $n = 20$:

- (a) $Y = X + \varepsilon$, $\varepsilon \sim N(0, 1)$, $X \sim U(0, 10)$
- (b) $Y = X^2 + \varepsilon$, $\varepsilon \sim N(0, 5)$, $X \sim U(-3, 3)$
- (c) $Y = e^X + \varepsilon$, $\varepsilon \sim N(0, 1)$, $X \sim U(0, 2)$

For each model, compute Pearson's r , Spearman's ρ , and Kendall's τ . Repeat 1000 times and compare the empirical power to detect association. What do you conclude about the robustness of each measure to non-linearity?

Exercise 13.8 (Conceptual Questions). Answer briefly:

- (a) Why are rank-based measures of association robust to outliers?
- (b) What is the difference between linear and monotonic association?
- (c) Under what conditions would you prefer Kendall's τ over Spearman's ρ ?
- (d) Explain the relationship between Spearman's ρ and Pearson's r applied to ranks.
- (e) How do ties affect Spearman's ρ and Kendall's τ ?
- (f) When should Fisher's exact test be used instead of the chi-square test for a contingency table?

A Statistical Tables

A.1 Critical Values for Spearman's Rank Correlation Coefficient

Table 1: Critical Values for Spearman's ρ (Two-Sided)

n	$\alpha = 0.10$		$\alpha = 0.05$		$\alpha = 0.01$	
	Lower	Upper	Lower	Upper	Lower	Upper
5	-1.000	1.000	-1.000	1.000	-	-
6	-0.886	0.886	-0.886	0.886	-0.943	0.943
7	-0.786	0.786	-0.786	0.786	-0.893	0.893
8	-0.738	0.738	-0.738	0.738	-0.833	0.833
9	-0.700	0.700	-0.700	0.700	-0.783	0.783
10	-0.648	0.648	-0.648	0.648	-0.745	0.745
11	-0.618	0.618	-0.618	0.618	-0.709	0.709
12	-0.587	0.587	-0.587	0.587	-0.678	0.678
13	-0.560	0.560	-0.560	0.560	-0.648	0.648
14	-0.538	0.538	-0.538	0.538	-0.622	0.622
15	-0.521	0.521	-0.521	0.521	-0.604	0.604
16	-0.503	0.503	-0.503	0.503	-0.585	0.585
17	-0.485	0.485	-0.485	0.485	-0.566	0.566
18	-0.472	0.472	-0.472	0.472	-0.550	0.550
19	-0.460	0.460	-0.460	0.460	-0.535	0.535
20	-0.447	0.447	-0.447	0.447	-0.522	0.522

A.2 Critical Values for Kendall's Tau

Table 2: Critical Values for Kendall's τ (Two-Sided)

n	$\alpha = 0.05$		$\alpha = 0.01$	
	Lower	Upper	Lower	Upper
5	-1.000	1.000	-1.000	1.000
6	-0.867	0.867	-0.867	0.867
7	-0.714	0.714	-0.810	0.810
8	-0.643	0.643	-0.714	0.714
9	-0.556	0.556	-0.667	0.667
10	-0.511	0.511	-0.600	0.600

Table 2 – continued from previous page

n	$\alpha = 0.05$		$\alpha = 0.01$	
	Lower	Upper	Lower	Upper
11	-0.491	0.491	-0.564	0.564
12	-0.455	0.455	-0.545	0.545
13	-0.436	0.436	-0.513	0.513
14	-0.407	0.407	-0.473	0.473
15	-0.390	0.390	-0.467	0.467
16	-0.367	0.367	-0.433	0.433
17	-0.353	0.353	-0.426	0.426
18	-0.340	0.340	-0.412	0.412
19	-0.322	0.322	-0.392	0.392
20	-0.316	0.316	-0.379	0.379

A.3 Critical Values for the Chi-Square Distribution

Table 3: Critical Values for the Chi-Square Distribution

df	$\chi^2_{0.10}$	$\chi^2_{0.05}$	$\chi^2_{0.025}$	$\chi^2_{0.01}$	$\chi^2_{0.001}$
1	2.706	3.841	5.024	6.635	10.828
2	4.605	5.991	7.378	9.210	13.816
3	6.251	7.815	9.348	11.345	16.266
4	7.779	9.488	11.143	13.277	18.467
5	9.236	11.070	12.833	15.086	20.515
6	10.645	12.592	14.449	16.812	22.458
7	12.017	14.067	16.013	18.475	24.322
8	13.362	15.507	17.535	20.090	26.124
9	14.684	16.919	19.023	21.666	27.877
10	15.987	18.307	20.483	23.209	29.588

B References

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Handwritten calculation for the chi-square test of independence:

	A	B	C	D	
X	30	40	20	60	Eng
Y	25	35	30	40	Math
	C	A	B	D	
X	20	30	40	60	
Y	2	1	3	4	

Calculation of the chi-square statistic:

$$\chi^2 = \frac{\sum \frac{(O - E)^2}{E}}{n(n-1)}$$

Where O is the observed frequency and E is the expected frequency.

Expected frequencies (E) are calculated as:

$$E_{AB} = \frac{(30+40)(25+35)}{100} = 12$$

$$E_{AC} = \frac{(30+20)(25+30)}{100} = 12$$

$$E_{AD} = \frac{(30+60)(25+30)}{100} = 12$$

$$E_{BC} = \frac{(40+20)(25+30)}{100} = 12$$

$$E_{BD} = \frac{(40+60)(25+30)}{100} = 12$$

$$E_{CD} = \frac{(20+60)(25+30)}{100} = 12$$

Chi-square calculation:

$$\chi^2 = \frac{(30-12)^2}{12} + \frac{(40-12)^2}{12} + \frac{(20-12)^2}{12} + \frac{(60-12)^2}{12} + \frac{(25-12)^2}{12} + \frac{(35-12)^2}{12} + \frac{(30-12)^2}{12} + \frac{(40-12)^2}{12} = 12$$

Final result:

$$\chi^2 = 12$$